

## 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

CS XI

Chapter 1: Computer Systems

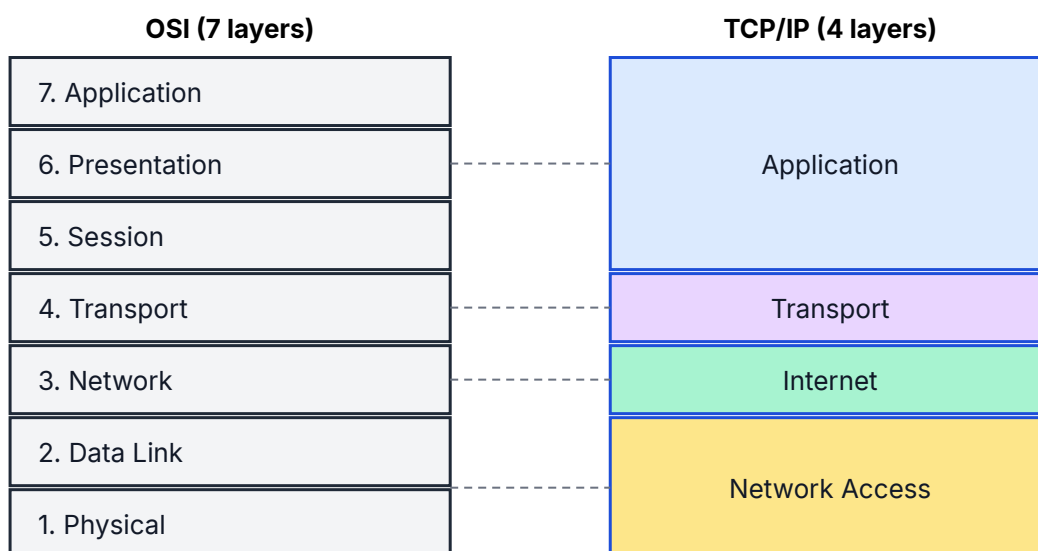
Lecture 12 of 12

### 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

#### LEARN IT

The OSI model is mainly a **reference** model used for teaching and design. The **TCP/IP model** is the **practical model used on the real Internet**. It condenses the 7 OSI layers into just **4 layers**.

| TCP/IP model (4 layers) | Corresponds to OSI layers                    |
|-------------------------|----------------------------------------------|
| 1. Application          | Application, Presentation, Session (7, 6, 5) |
| 2. Transport            | Transport (4)                                |
| 3. Internet             | Network (3)                                  |
| 4. Network Access       | Data Link, Physical (2, 1)                   |



How the 7 OSI layers map onto the 4 TCP/IP layers.

| Point            | OSI model | TCP/IP model |
|------------------|-----------|--------------|
| Number of layers | 7         | 4            |

| Use           | Reference / teaching model                            | Practical model used on the real Internet     |
|---------------|-------------------------------------------------------|-----------------------------------------------|
| Top layers    | Separate Application, Presentation and Session layers | All three combined into one Application layer |
| Bottom layers | Separate Data Link and Physical layers                | Combined into one Network Access layer        |

#### WORKED EXAMPLE

##### Where does each protocol/device go in TCP/IP?

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

#### REMEMBER IT

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total  $3+1+1+2 = 7$ .

#### COMMON MISTAKE

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

#### اردو میں سمجھیں

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

#### CHECK YOURSELF

1. How many layers does the TCP/IP model have?
2. Which TCP/IP layer corresponds to OSI layers 2 and 1?
3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** OSI has:

- a. 7 layers
- b. 4 layers
- c. 2 layers
- d. 12 layers

**MCQ 2.** Layer 7 is:

- a. Application
- b. Physical
- c. Network
- d. Data link

**MCQ 3.** IP addressing lives at:

- a. Network (layer 3)
- b. Application
- c. Physical
- d. Session

**MCQ 4.** Cables and bits are:

- a. Physical
- b. Presentation
- c. Session
- d. Application

**MCQ 5.** End-to-end ports (TCP/UDP) are:

- a. Transport
- b. Network
- c. Physical
- d. Data link

**MCQ 6.** The model that explains “how data travels” is a:

- a. Communication model
- b. Sort algorithm
- c. K-map
- d. MVP

## Extra short questions

1. Write the 7 OSI layers from top to bottom.
2. One job of Transport and of Network.
3. Why layer models?
4. Mnemonic you will actually write in the exam.

## Extra long question

1. Name all 7 OSI layers with one job and one protocol/hardware example each. ★  
Golden.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Application, Presentation, Session, Transport, Network, Data Link, Physical.
2. Transport: reliability/ports (TCP/UDP). Network: routing and logical addressing (IP).
3. Each layer has a small job; vendors can change one layer without rewriting the rest.
4. Please Do Not Throw Sausage Pizza Away (bottom-up) or All People Seem To Need Data Processing (top-down).

**Long**

1. 7 App HTTP/DNS; 6 Pres encryption/JPEG; 5 Sess dialogs; 4 Trans TCP/UDP; 3 Net IP/routers; 2 DL frames/MAC/switches; 1 Phy cables/bits/hubs. Draw the stack.

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Next lecture: Chapter exam (in the chapter lecture PDF)